Predicting Parole Violators

In many criminal justice systems around the world, inmates deemed not to be a threat to society are released from prison under the parole system prior to completing their sentences. They are still considered to be serving their sentences while on parole, and they can be returned to prison if they violate the terms of their parole.

Parole boards are charged with identifying which inmates are good candidates for release on parole. They seek to release inmates who will not commit additional crimes after release. In this problem, we will build and validate a model that predicts if an inmate will violate the terms of his or her parole. Such a model could be useful to a parole board when deciding to approve or deny an application for parole.

The file *Parole.csv* contains data from 2004 on parolees who served no more than 6 months in prison and whose maximum sentence for all charges did not exceed 18 months. All parolees in the dataset either successfully completed the term of their parole during 2004 or violated the terms of their parole in 2004. The variables in this dataset are described in Table 22.5.

Table 22.5: Variables in the dataset *Parole.csv*.

Variable	Description
`Male`	1 if the parolee is male, and 0 if female.
`RaceWhite`	1 if the parolee is white, and 0 otherwise.
`Age`	The parolee's age in years when he or she was released from prison.
`State`	The parolee's state, either Kentucky, Louisiana, Virginia, or Other. These three states were selected due to having a high representation in the dataset.
`TimeServed`	The number of months the parolee served in prison (limited by the inclusion criteria to not exceed 6 months).
`MaxSentence`	The maximum sentence length for all charges, in months (limited by the inclusion criteria to not exceed 18 months).
`MultipleOffenses`	1 if the parolee was incarcerated for multiple offenses, and 0 otherwise.
`Crime`	The parolee's main crime leading to incarceration, either Larceny, Drugs (representing a drug-related crime), Driving (representing a driving-related crime), or Other.
`Violator`	1 if the parolee violated their parole, and 0 if the parolee completed their parole without violation.

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Using this dataset, answer the following questions.

a) How many parolees do we have data for? Of the parolees that we have data for, what percentage violated the terms of their parole?

b) Randomly split the data into a training set and a testing set, putting 70% of the data in the training set. Then, build a logistic regression model to predict the variable `Violator` using all of the other variables as independent variables. You should use the training dataset to build the model.

i) Describe your resulting model. Which variables are significant in your model?

ii) Consider a parolee who is male, of white race, aged 50 years at prison release, from the state of Maryland, served 3 months, had a maximum sentence of 12 months, did not commit multiple offenses, and committed a larceny. According to your model, what is the probability that this individual is a violator? (HINT: You should use the coefficients of your model and the Logistic Response Function to solve this problem.)

iii) Now compute the model's predicted probabilities for parolees in the testing set. Then create a confusion matrix for the test set using a threshold of 0.5. What is the model's false positive rate on the test set? False negative rate? Overall accuracy?

iv) Compare your accuracy on the test set to a *baseline* model that predicts every parolee in the test set is a non-violator, regardless of the values of the independent variables. Does your model improve over this simple model?

v) Consider a parole board that might use your model to predict whether parolees will be violators or not. The job of a parole board is to make sure that a prisoner is ready to be released into free society, and therefore parole boards tend to be particularly concerned about releasing prisoners who will violate their parole. Would the parole board be more concerned by false positive errors or false negative errors? How should they adjust their threshold to reflect their error preferences?

vi) Compute the AUC of the model on the test set, and interpret what the number means in this context. Considering the AUC, the accuracy compared to the baseline model, and what happens when the threshold is adjusted, do you think this model is of value to a parole board? Why or why not?

c) Our goal in this problem has been to predict the outcome of a parole decision, and we used a publicly available dataset of parole releases for predictions. It is always important to evaluate a dataset for possible

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sources of *bias*, especially when the dataset only contains a subset of the observations of interest.

The *Parole.csv* dataset contains all individuals released from parole in 2004, either due to completing their parole term or violating the terms of their parole. However, it does not contain parolees who neither violated their parole nor completed their term in 2004, causing non-violators to be underrepresented. This is called *selection bias* or *selecting on the dependent variable*, because we used our dependent variable (parole violation) to select only a subset of all relevant parolees to include in our analysis. How could we improve our dataset to best address selection bias?

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